

Chapter 10

Conclusion and Key Takeaways

Successful traders do not rely on luck. Rather, the long-term success of traders depends on their ability to obtain a consistent, statistical edge from the tools, strategies, and information available. This book introduces the core concepts of options trading and teaches new traders how to capitalize on the versatility and capital efficiency of options in a personalized and responsible way. Options are fairly complicated instruments, but this book aims to lessen the learning curve by focusing on the most essential aspects of applied options trading. The detailed framework laid out in this book can be summarized succinctly in the following key takeaways:

1. Implied volatility (IV) is a proxy for the sentiment of market risk derived from supply and demand for financial insurance. When options prices increase, IV increases; when options prices decrease, IV decreases. IV gives the perceived magnitude of future movements and is not directional. It can also be used to approximate the one standard deviation expected price range of an asset (although this does not take strike skew into account). The CBOE Volatility Index (VIX) is meant to track the IV for the S&P 500 and is used as a proxy for the perceived risk of the broader market. The VIX, like all volatility signals, is assumed to revert down following significant expansions, which indicates some statistical validity in making downward directional assumptions about volatility once it is inflated.
2. Compared to long premium strategies, short premium strategies yield more consistent profits and have the long-term statistical advantage. The trade-off for receiving consistent profits is exposure to large and sometimes undefined losses, which is why the two most important goals of a short premium trader are to profit consistently enough to cover moderate, more likely losses and to construct a portfolio that can survive unlikely extreme losses.
3. The profitability of short options strategies depends on having a large number of occurrences to reach positive statistical averages, a

consequence of the law of large numbers and the central limit theorem. At minimum, approximately 200 occurrences are needed for the average profit and loss (P/L) of a strategy to converge to long-term profit targets and more is better.

4. Extreme losses for short premium positions are highly unlikely but typically happen when price swings in the underlying are large while the expected move range is tight (low IV). Because large price movements in low IV are rare and difficult to reliably model, the most effective way to reduce this exposure is to trade short premium once IV is elevated.
5. Although high volatility environments are ideal for short premium positions, short premium positions have high probability of profits (POPs) and some statistical edge in all volatility environments. Additionally, because volatility is low the majority of the time, trading short options strategies in *all* IV environments allows traders to profit more consistently and increases the number of occurrences. To manage exposure to outlier risk when adopting this strategy, it is essential to maintain small position sizes and limit the amount of capital allocated to short premium positions, especially when IV is low. This strategy can be further improved by scaling the amount of capital allocated to short premium according to the current market conditions.

VIX Range	Maximum Portfolio Allocation
0–15	25%
15–20	30%
20–30	35%
30–40	40%
40+	50%

6. Buying power reduction (BPR) is the amount of portfolio capital required to place and maintain an option trade. The BPR for long options is merely the cost of the contract, and the BPR for short options is meant to encompass at least 95% of the potential losses for exchange-traded fund (ETF) underlyings and 90% of the potential losses for stock underlyings. BPR is used to evaluate short premium risk on a trade-by-trade basis in two ways: BPR is a fairly reliable

metric for the worst-case loss of an undefined risk position, and BPR is used to determine if a position is appropriate for a portfolio based on its buying power. A defined risk trade should not occupy more than 5% of portfolio buying power and an undefined risk trade should not occupy more than 7%, with exceptions allowed for small accounts. The formulae for BPR are complicated and specific to the type of strategy, but the BPR for short strangles is approximately 20% of the price of the underlying. BPR can be used to compare the risk for variations of the same strategy (e.g., strangle on underlying A versus strangle on underlying B), but it cannot be used to compare risk for strategies with different risk profiles (e.g., strangle on underlying A versus iron condor on underlying A).

7. Traders trade according to their personal profit goals, risk tolerances, and market beliefs, but it is generally good practice to be aware of the following:

- Only trade underlyings with liquid options markets to minimize illiquidity risk.
- The choice of underlying is somewhat arbitrary, but it's important to select an underlying with an appropriate level of risk. Stock underlyings tend to be higher-risk, higher-reward than ETF underlyings. This means stock underlyings present high IV opportunities more frequently, but they have more tail loss exposure and are more expensive to trade.
- Choose a contract duration that is an efficient use of buying power, allows for consistency, offers a reasonable number of occurrences, has manageable P/L swings throughout the duration, and has moderate ending P/L variability. Durations between 30 and 60 days are suitable for most traders.
- Compared to defined risk trades, undefined risk trades have higher POPs, higher profit potentials, unlimited downside risk, and higher BPRs. High-POP defined risk trades, such as wide iron condors, have comparable risk profiles to undefined risk trades while also offering protection from extreme losses. Such trades can be better suited for low IV conditions compared to

undefined risk trades and are allowed to occupy undefined risk portfolio capital.

- Contracts with higher deltas are higher-risk, higher-reward than contracts with lower deltas. When trading premium, consider contracts between 10Δ and 40Δ , which is large enough to achieve a reasonable amount of growth but small enough to have manageable P/L swings and ending P/L variability.

8. When choosing a management strategy, the primary factors to consider are convenience and consistency, capital allocation preferences, desired number of occurrences, per-trade average P/L, and per-trade exposure. Early-managed positions have lower per-trade P/Ls but less tail risk than positions held to expiration. Because managing early also accommodates more occurrences and averages more P/L per day, closing positions prior to expiration and redeploying capital to new positions is generally a more efficient use of capital compared to extracting more value from an existing position.

- If managing according to days to expiration (DTE), consider closing trades around the contract duration midpoint to achieve a decent amount of long-term profit and justify the tail loss exposure.
- If managing an undefined position according to a profit target, choosing a target between 50% and 75% of the initial credit allows for reasonable profits while also reducing the potential magnitude of outlier losses. Choosing a profit target that is too low reduces average P/L, and choosing a profit target that is too high does little to mitigate outlier risk. Profit targets for defined risk positions can be lower because they are generally less volatile.
- If combining strategies, managing undefined risk contracts at either 50% of the initial credit or halfway through the contract duration generally achieves reasonable, consistent profits and moderates outlier risk.
- If implementing a stop loss, a mid-range stop loss threshold of at least -200% of the initial credit is practical. If the stop loss is too

small (–50% for example), losses are more likely since options have significant P/L variance, although they often recover. It's also important to note that stop losses do not guarantee a maximum loss in cases of rapid price movements, so stop losses are typically paired with another management strategy unless trading passively. Stop losses are generally not suitable for defined risk strategies.

9. Maintaining the capital allocation guidelines is crucial for limiting tail exposure and achieving a reasonable amount of long-term growth:

- The amount of portfolio buying power allotted to short premium positions, such as short strangles and short iron condors, should range from 25% to 50% depending on the current market volatility, with the remaining capital either kept in cash or a low-risk passive investment. [refer to Takeaway 5].
- Of the amount allocated to short premium, at least 75% should be reserved for undefined risk trades (with less than 7% of portfolio buying power allocated to a single position) and at most 25% reserved for defined risk strategies (with less than 5% of portfolio buying power allocated to a single position) [refer to Takeaway 6].
- Generally speaking, at most 25% of the capital allocated to short premium should go toward supplemental positions, or higher-risk, higher-reward trades that are tools for market engagement. The remainder should go toward core positions or trades with high POPs and moderate P/L variation that offer consistent profits and reasonable outlier exposure.

10. Diversifying the underlyings of an options portfolio (i.e., trading a collection of assets with low correlations) is one of the most essential diversification tools for portfolio risk management, particularly outlier risk management. Strategy diversification and duration diversification, though not as essential as underlying diversification, are other straightforward risk management techniques.

11. The Greeks form a set of risk measures that quantify different dimensions of exposure for options. Each contract has its own specific

set of Greeks, but some Greeks are additive across positions with different underlyings. Consequently, these metrics can be used to summarize the various sources of risk for a portfolio and guide adjustments. Two key Greeks are beta-weighted delta ($\beta\Delta$) and theta (θ). Beta-weighted delta represents the amount of directional exposure a position has relative to some index rather than the underlying itself. Theta represents the expected decrease in an option's value per day. $\beta\Delta$ neutral portfolios are attractive to investors because they are relatively insensitive to directional moves in the market and profit from changes in IV and time.

12. Because short-premium traders consistently profit from time decay, the total theta across positions gives a reliable estimate for the expected daily growth of a short options portfolio. The minimum theta ratio ($\theta_{\text{portfolio}}/\text{net liquidity}$) for an options portfolio should range from 0.05% to 0.1% and should not exceed 0.2% because this indicates excessive risk. If a portfolio is not meeting these theta ratio guidelines, then the positions should be adjusted as follows:

- If a properly allocated, well-diversified portfolio is $\beta\Delta$ neutral but does not provide a sufficient amount of theta, then the positions in the portfolio should be reevaluated. In this case, perhaps some defined risk trades should be replaced with undefined risk trades or undefined risk positions be rolled to higher deltas. New delta neutral positions can also be added, such as strangles and iron condors, for example. IV and theta are also highly correlated, meaning that higher IV underlyings could also be considered if theta is too low. These measures can be reversed if the portfolio has too much theta exposure while being $\beta\Delta$ neutral.
- If the theta ratio is too low ($<0.1\%$) and the portfolio is not $\beta\Delta$ neutral, then either existing positions should be re-centered or tightened or new short premium positions should be added.

If the $\beta\Delta$ is too large and positive (bullish), then add new negative $\beta\Delta$ positions (e.g., add short calls on positive beta

underlyings or add short puts on negative beta underlyings).

If the $\beta\Delta$ is too large and negative (bearish), then add new positive $\beta\Delta$ positions (e.g., add short puts on positive beta underlyings).

- If the theta ratio is too large ($>0.2\%$) and the portfolio is not $\beta\Delta$ neutral, then either existing positions should be re-centered or widened or short premium positions should be removed.

If the $\beta\Delta$ is too large and positive (bullish), then remove positive $\beta\Delta$ positions (e.g., remove short puts on positive beta underlyings).

If the $\beta\Delta$ is too large and negative (bearish), then remove negative $\beta\Delta$ positions (e.g., remove short calls on positive beta underlyings).

- If a properly allocated, well-diversified portfolio provides a sufficient amount of theta but is not $\beta\Delta$ neutral, then existing positions should be reevaluated. For example, skewed positions could be closed and re-centered or replaced with new delta-neutral positions that offer comparable amounts of theta.

13. Binary event trades, such as trades around quarterly earnings reports, should be traded cautiously, only occupy spare portfolio capital, and their position size should be kept exceptionally small. Binary event trades must be carefully monitored and typically take place over much shorter timescales than more typical trades. They are often opened the day before a binary event and closed the day after.

Options trading is not for everyone. However, for traders who are prepared to understand the complex risk profiles of options, comfortable accepting a certain level of exposure, and willing to commit the time to active trading, short premium strategies can offer a probabilistic edge and the potential to profit in any type of market. There is no “right” way to trade these instruments; all traders have unique profit goals and risk tolerances. It is our hope that this book will guide traders to make informed decisions that best align with their personal objectives.

OceanofPDF.com